

Customer Churn Analysis

Python | PostgreSQL | Power BI

GitHub: [Project Link](#)

Dashboard: [Power BI Dashboard](#)

Project Overview → 93,587 Customers from India and Nepal; Subscriptions started 2018 to 2024, Cancellations recorded up to August 2026.

Result → 19.42% Churn Rate, 395.64K Monthly Revenue Lost, 486.97K Monthly Revenue still at high risk.

1- Business Problem

A subscription-based service provider is facing challenges with customer retention. They receive thousands of complaints and don't know why customers leave or whether complaint handling helps. With no new sign-ups since December 2024. They are currently reacting to cancellations after they happen. They want to analyze the data to understand:

- **Who** is leaving, and whether any plan, country, gender, age group or risk segment stands out.
- **When** they leave, so they know the right time to step in.
- **Why** they leave, so they can fix the real causes instead of guessing.
- **How much revenue** is affected, so they can decide where to spend retention budget.

2- Dataset Summary

Three source tables (db_customer, db_subscription, db_support) were cleaned and merged into one table (Cleaned_Merged_Data) with 93,587 rows and 25 columns. Source of the dataset: [\[Cleaned Data\]](#).

3-Python-Data Cleaning & Feature Engineering

Python (Pandas) was used to clean the three tables before merging them into one:

- **Customer table:** Dropped mostly empty columns, removed 1,000 duplicates, and standardized gender, country, state and name formats.
- **Subscription table:** Fixed spelling errors, date formats, the \$ sign, negative values, out-of-range scores and illogical dates.
- **Support table:** Dropped the empty column, removed duplicates, and standardized escalations (Y/N → Yes/No).
- **All tables:** Handled null values and fixed data types.

4)- SQL (PostgreSQL): Analysis and KPIs

- The data was cleaned in Python, so no cleaning was done in SQL. The cleaned table (Cleaned_Merged_Data) was exported from Python and imported into PostgreSQL, where it was used for analysis.
- SQL queries were written to answer the business questions. Each query was checked against the dashboard numbers, such as a churn rate of 19.42% and 18,176 churned customers.

5)- Power BI Dashboard

The report has 6 pages with a navigation panel, slicers and a reset button.

- Home: KPI cards, filters, key review points and recommendations.
- Customer Segmentation: Customers by risk band, plan, age and gender.
- Churn Overview: Cancellation reasons, churn by plan, yearly churn trend.
- Risk & Complaints: Churn by risk band and complaint count, escalations.
- Revenue Impact: ARPU, CLTV, revenue lost and revenue at high risk.
- Time Trends: Churn and revenue lost by year; revenue at risk by start year.

6)- Key Insights

- Churn rate is **19.42%** (18,176 of 93,587 customers).
- Revenue lost is **395.6K**, with **487K** still at risk monthly.
- High Risk customers churn at 37.5% and cause about **287.16K** of lost revenue.
- Cancellation reasons are evenly spread. Service, billing and content issues make about 50%.
- Plan, gender and age show no real difference.
- Complaints don't predict churn (about 19–20% at every complaint count).
- Churn is flat at about **2.6–2.7K** a year from 2020 to 2024. The 2026 figure is partial.

7)- Business Recommendations

- Offer **retention deals** such as discounts, free upgrades or loyalty perks to High-Risk customers first.
- Fix service, billing and streaming quality with better support.
- Run win-back offers, starting with competitor switchers especially for customers who left for price or competitor reasons.
- Track the **risk score** monthly and review how complaints are resolved.
- Resume **customer acquisition**, since there are no new sign-ups after December 2024.